



THE UNIVERSITY
OF QUEENSLAND

A U S T R A L I A

This exam paper must not be removed from the venue

Venue

Seat Number

Student Number

Family Name

First Name

School of Mathematics & Physics

Semester Two Examinations, 2024

MATH1061 Discrete Mathematics

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

•No written or printed material permitted

•Casio FX82 series or UQ approved and labelled calculator only

•During Planning Time - Students are encouraged to review and plan responses to the exam questions

Materials Permitted in the Exam Venue:

(No electronic aids are permitted e.g. laptops, phones)

None

Materials to be supplied to Students:

Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

None

Instructions to Students:

If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

Answer all questions in the space provided. Show all working; answers given without justification may not receive full marks. Questions carry the number of marks shown.

Total marks available: 85

For Examiner Use Only

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
Total _____	

1. (a) For statement variables p , q and r , prove that

$$[(p \vee \sim q) \wedge (q \vee r)] \rightarrow (p \vee r)$$

is a tautology.

You may use either truth tables or logical equivalences. If you use logical equivalences, you should name the laws you use at each step.

(4 marks)

- (b) Find values of the statement variables a , b and c that show that the following argument is invalid. You do not need to show your working.

1. $a \rightarrow b$
 2. $\sim a \rightarrow c$
 3. $\sim b \rightarrow c$
- $\therefore b$

(2 marks)

2. (a) Prove the following statement:

For all $a, b \in \mathbb{N}$, if a is a perfect square and b is not a perfect square then $a \times b$ is not a perfect square.

Recall that $\mathbb{N} = \{1, 2, 3, \dots\}$ represents the natural numbers, and a perfect square is a number of the form x^2 where x is an integer.

Hint: You might find unique prime factorisation to be helpful.

(4 marks)

2. (b) Find a counterexample that disproves the following statement:

For all $x \in \mathbb{R}$, $\lfloor x^2 \rfloor = \lfloor x \rfloor^2$.

You do not need to show your working.

Recall that $\lfloor \cdot \rfloor$ denotes the floor function.

(2 marks)

2. (c) Using the Euclidean algorithm, compute $\gcd(116, 88)$. Show your working.

(3 marks)

- (d) Using your answer to part (c), compute $\text{lcm}(116, 88)$.

(2 marks)

3. Let $a_0, a_1, a_2, \dots$ be the sequence defined recursively by $a_0 = 0$, $a_1 = 1$, and

$$a_n = 3a_{n-1} - 2a_{n-2} \text{ for all integers } n \geq 2.$$

- (a) Compute the terms a_2 , a_3 and a_4 .

(3 marks)

- (b) Guess an explicit formula for this sequence, and prove that your formula is correct.

(Hint: If you have trouble guessing, you may find that computing more terms can help.)

(4 marks)

4. (a) Write out the elements of $\{1, 2\} \cap \{1, 3\}$.

(1 mark)

- (b) Write out the elements of $\{1, 2\} \cup \{1, 3\}$.

(1 mark)

- (c) Write out the elements of $\{1, 2\} \times \{1, 3\}$.

(2 marks)

- (d) Write out the elements of $\mathcal{P}(\{1, 2\})$.

(2 marks)

- (e) What is the cardinality of $\mathcal{P}(\mathcal{P}(\{1, 2\}))$? You should write your answer as a single integer. You do not need to show your working.

(2 marks)

5. Consider the function $f : \mathbb{R} \rightarrow \mathbb{Z}$ defined by $f(x) = \lfloor x^2 \rfloor$.

Recall that $\lfloor \cdot \rfloor$ denotes the floor function.

- (a) What is the image of the interval $[-1, 1]$ under the function f ?
You do not need to show your working.

(2 marks)

- (b) What is the preimage of 1 under the function f ?
You do not need to show your working.

(2 marks)

- (c) Is f one-to-one?
You should justify your answer with an explanation.

(3 marks)

- (d) Is f onto?
You should justify your answer with an explanation.

(3 marks)

6. (a) Prove that the interval $[0, 1]$ has the same cardinality as the set $\mathbb{R}_+$ of all positive real numbers.

(Hint: You might wish to use the Schröder-Bernstein theorem.)

(5 marks)

6. (b) Does $|\mathbb{Z}| = |\mathbb{Q}|$? You do not need to show your working.

(1 mark)

- (c) Does $|\mathbb{Z}| = |\mathbb{Z} \times \mathbb{Z}|$? You do not need to show your working.

(1 mark)

- (d) Does $|\mathbb{R}| = |\mathbb{Q} \times \mathbb{Q}|$? You do not need to show your working.

(1 mark)

7. Define a relation ρ on $\mathbb{R}$ by $x\rho y$ if and only if $x \times y > 0$.

(a) Is ρ reflexive? Explain why or why not.

(2 marks)

(b) Is ρ symmetric? Explain why or why not.

(2 marks)

7. As before, define a relation ρ on $\mathbb{R}$ by $x\rho y$ if and only if $x \times y > 0$.

(c) Is ρ transitive? Explain why or why not.

(2 marks)

(d) Is ρ antisymmetric? Explain why or why not.

(2 marks)

8. (a) What is the coefficient of x^2 in the expansion of $(3 - x)^6$? You should give your answer as a single integer, and show your working.

(2 marks)

- (b) How many one-to-one functions are there from $\{a, b, c\}$ to $\{1, 2, 3, 4, 5\}$? You should give your answer as a single integer, and show your working.

(3 marks)

8. (c) How many onto functions are there from $\{1, 2, 3, 4, 5\}$ to $\{a, b, c\}$? You should give your answer as a single integer, and show your working.

(5 marks)

9. Let G denote the group $(\mathbb{Z}_2 \times \mathbb{Z}_2, +)$. Here we have $(a, b) + (c, d) = (a + c, b + d)$, where addition is done modulo 2.

Let H denote the group $(\mathbb{Z}_5 - \{0\}, \times)$. Here multiplication is done modulo 5.

Note that both G and H contain exactly four elements.

- (a) Write out the full Cayley tables for G and H .

(4 marks)

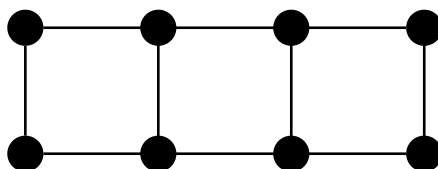
- (b) Are the groups G and H isomorphic? Explain why or why not.

(3 marks)

10. (a) A graph has 10 edges and 6 vertices. Five of the vertices have degree 3. What is the degree of the remaining vertex?

(3 marks)

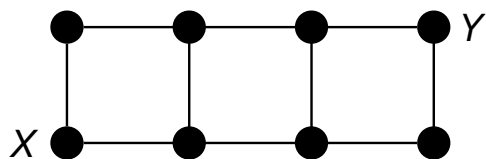
- (b) Consider the graph illustrated below. Does this graph have an Eulerian trail? You should justify your answer with an explanation.



(3 marks)

10. (c) Consider again the graph illustrated below. Prove that there is no walk of odd length that starts at vertex X and ends at vertex Y .

Recall that the length of a walk is the number of edges that it traverses.



(4 marks)

END OF EXAMINATION

MATH1061/MATH7861 Examination Formula Pages

Logical Equivalences

Given any statement variables p , q and r , the following logical equivalences hold, where **t** denotes a tautology and **c** denotes a contradiction:

<i>Commutative laws</i>	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
<i>Associative laws</i>	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
<i>Distributive laws</i>	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
<i>Identity laws</i>	$p \wedge \mathbf{t} \equiv p$	$p \vee \mathbf{c} \equiv p$
<i>Negation laws</i>	$p \vee \sim p \equiv \mathbf{t}$	$p \wedge \sim p \equiv \mathbf{c}$
<i>Double negative laws</i>	$\sim(\sim p) \equiv p$	
<i>Idempotent laws</i>	$p \wedge p \equiv p$	$p \vee p \equiv p$
<i>Universal bound laws</i>	$p \vee \mathbf{t} \equiv \mathbf{t}$	$p \wedge \mathbf{c} \equiv \mathbf{c}$
<i>De Morgan's laws</i>	$\sim(p \wedge q) \equiv \sim p \vee \sim q$	$\sim(p \vee q) \equiv \sim p \wedge \sim q$
<i>Absorption laws</i>	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
<i>Negations of t and c</i>	$\sim \mathbf{t} \equiv \mathbf{c}$	$\sim \mathbf{c} \equiv \mathbf{t}$

Valid Argument Forms

Modus Ponens	$p \rightarrow q$ p $\therefore q$	Elimination	(a) $p \vee q$ $\sim q$ $\therefore p$	(b) $p \vee q$ $\sim p$ $\therefore q$
Modus Tollens	$p \rightarrow q$ $\sim q$ $\therefore \sim p$	Transitivity	$p \rightarrow q$ $q \rightarrow r$ $\therefore p \rightarrow r$	
Generalization	(a) p $\therefore p \vee q$	Proof by division into cases	$p \vee q$ $p \rightarrow r$ $q \rightarrow r$ $\therefore r$	(b) q $\therefore p \vee q$
Specialization	(a) $p \wedge q$ $\therefore p$			(b) $p \wedge q$ $\therefore q$
Conjunction	p q $\therefore p \wedge q$	Contradiction Rule	$\sim p \rightarrow \mathbf{c}$ $\therefore p$	

Set Identities

Given any sets A , B and C that are subsets of a universal set U , the following laws hold:

- Commutative Laws (a) $A \cup B = B \cup A$ (b) $A \cap B = B \cap A$
- Associative Laws (a) $(A \cup B) \cup C = A \cup (B \cup C)$
(b) $(A \cap B) \cap C = A \cap (B \cap C)$
- Distributive Laws (a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- Identity Laws (a) $A \cup \emptyset = A$ (b) $A \cap U = A$
- Complement Laws (a) $A \cup A^c = U$ (b) $A \cap A^c = \emptyset$
- Double Complement Law $(A^c)^c = A$
- Idempotent Laws (a) $A \cup A = A$ (b) $A \cap A = A$
- Universal Bound Laws (a) $A \cup U = U$ (b) $A \cap \emptyset = \emptyset$
- De Morgan's Laws (a) $(A \cup B)^c = A^c \cap B^c$ (b) $(A \cap B)^c = A^c \cup B^c$
- Absorption Laws (a) $A \cup (A \cap B) = A$ (b) $A \cap (A \cup B) = A$
- Complement of U and $\emptyset$ (a) $U^c = \emptyset$ (b) $\emptyset^c = U$
- Set Difference Law $A - B = A \cap B^c$

The Quotient-Remainder Theorem Given any integer n and any positive integer d , there exist unique integers q and r such that $n = dq + r$ and $0 \leq r < d$.

Unique Factorisation Theorem Given any integer $n > 1$, there exists a positive integer k , distinct prime numbers $p_1, p_2, \dots, p_k$, and positive integers $e_1, e_2, \dots, e_k$ such that $n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$, and any other expression for n as a product of prime numbers is identical to this except perhaps for the order in which the factors are written.

Schröder-Bernstein Theorem For all sets A and B , if $|A| \leq |B|$ and $|B| \leq |A|$ then $|A| = |B|$.

Binomial Theorem Given any real numbers a and b , and any nonnegative integer n ,

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k.$$